MySQL Stored Procedure – Assignment

Q1. Check Even or Odd

Create a stored procedure check_even_odd that accepts an integer and displays whether the number is Even or Odd.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE check_even_odd(IN num INT)
    -> BEGIN
    ->   IF MOD(num, 2) = 0 THEN
    ->     SELECT CONCAT(num, ' is Even') AS Result;
    ->   ELSE
    ->     SELECT CONCAT(num, ' is Odd') AS Result;
    ->   END IF;
    -> END //
Query OK, 0 rows affected
mysql> DELIMITER ;
mysql> CALL check_even_odd(25);
+-----------+
| Result    |
+-----------+
| 25 is Odd |
+-----------+
1 row in set
mysql> CALL check_even_odd(40);
+------------+
| Result     |
+------------+
| 40 is Even |
+------------+
1 row in set
```

Q2. Calculate Simple Interest

Create calculate_interest accepting principal, rate of interest, and time in years. Simple Interest = (Principal x Rate x Time) / 100.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE calculate_interest(IN p DECIMAL(10,2), IN r DECIMAL(5,2), IN t INT)
    -> BEGIN
    ->   SELECT (p * r * t) / 100 AS SimpleInterest;
    -> END //
Query OK, 0 rows affected
mysql> DELIMITER ;
mysql> CALL calculate_interest(10000, 5, 2);
+----------------+
| SimpleInterest |
+----------------+
|        1000.00 |
+----------------+
1 row in set
mysql> CALL calculate_interest(25000, 7, 3);
+----------------+
| SimpleInterest |
+----------------+
|        5250.00 |
+----------------+
1 row in set
```

Q3. Calculate Grade

Create calculate_grade accepting student ID, student name, and marks. Grade: 90-100 A+, 80-89 A, 70-79 B, 60-69 C, 50-59 D, below 50 F.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE calculate_grade(IN sid INT, IN sname VARCHAR(50), IN marks INT)
    -> BEGIN
    ->   DECLARE grade VARCHAR(2);
    ->   IF marks >= 90 THEN SET grade = 'A+';
    ->   ELSEIF marks >= 80 THEN SET grade = 'A';
    ->   ELSEIF marks >= 70 THEN SET grade = 'B';
    ->   ELSEIF marks >= 60 THEN SET grade = 'C';
    ->   ELSEIF marks >= 50 THEN SET grade = 'D';
    ->   ELSE SET grade = 'F';
    ->   END IF;
    ->   SELECT sid AS StudentID, sname AS StudentName, marks AS Marks, grade AS Grade;
    -> END //
Query OK, 0 rows affected
mysql> DELIMITER ;
mysql> CALL calculate_grade(101, 'Amit', 85);
+-----------+-------------+-------+-------+
| StudentID | StudentName | Marks | Grade |
+-----------+-------------+-------+-------+
|       101 | Amit        |    85 | A     |
+-----------+-------------+-------+-------+
1 row in set
mysql> CALL calculate_grade(102, 'Priya', 72);
+-----------+-------------+-------+-------+
| StudentID | StudentName | Marks | Grade |
+-----------+-------------+-------+-------+
|       102 | Priya       |    72 | B     |
+-----------+-------------+-------+-------+
1 row in set
mysql> CALL calculate_grade(103, 'Rahul', 45);
+-----------+-------------+-------+-------+
| StudentID | StudentName | Marks |
+-----------+-------------+-------+
|       103 | Rahul       |    45 |
+-----------+-------------+-------+-------+
```

Q4. Calculate Employee Bonus

Create calculate_bonus accepting employee name and salary. Bonus rates: salary >= 1,000,000: 20%; >= 50,000: 15%; >= 30,000: 10%; below 30,000: 5%. Display employee name, salary, bonus percentage, bonus amount, and total salary.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE calculate_bonus(IN ename VARCHAR(50), IN sal DECIMAL(12,2))
    -> BEGIN
    ->   DECLARE pct DECIMAL(5,2);
    ->   IF sal >= 1000000 THEN SET pct = 20;
    ->   ELSEIF sal >= 50000 THEN SET pct = 15;
    ->   ELSEIF sal >= 30000 THEN SET pct = 10;
    ->   ELSE SET pct = 5;
    ->   END IF;
    ->   SELECT ename AS EmployeeName, sal AS Salary, pct AS BonusPercent,
    ->          sal*pct/100 AS BonusAmount, sal+(sal*pct/100) AS TotalSalary;
    -> END //
Query OK, 0 rows affected
mysql> DELIMITER ;
mysql> CALL calculate_bonus('Amit', 120000);
+--------------+-----------+--------------+-------------+-------------+
| EmployeeName | Salary    | BonusPercent | BonusAmount | TotalSalary |
+--------------+-----------+--------------+-------------+-------------+
| Amit         | 120000.00 |        15.00 |    18000.00 |   138000.00 |
+--------------+-----------+--------------+-------------+-------------+
1 row in set
mysql> CALL calculate_bonus('Priya', 65000);
+--------------+----------+--------------+-------------+-------------+
| EmployeeName | Salary   | BonusPercent | BonusAmount | TotalSalary |
+--------------+----------+--------------+-------------+-------------+
| Priya        | 65000.00 |        15.00 |     9750.00 |    74750.00 |
+--------------+----------+--------------+-------------+-------------+
1 row in set
mysql> CALL calculate_bonus('Rahul', 25000);
+--------------+----------+--------------+-------------+-------------+
| EmployeeName | Salary   | BonusPercent | BonusAmount | TotalSalary |
+--------------+----------+--------------+-------------+-------------+
| Rahul        | 25000.00 |         5.00 |     1250.00 |    26250.00 |
+--------------+----------+--------------+-------------+-------------+
1 row in set
```

Q5. Find Maximum of Three Numbers

Create find_maximum accepting three integers and displaying the largest number.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE find_maximum(IN a INT, IN b INT, IN c INT)
    -> BEGIN
    ->   SELECT GREATEST(a, b, c) AS MaximumNumber;
    -> END //
Query OK, 0 rows affected
mysql> DELIMITER ;
mysql> CALL find_maximum(25, 78, 45);
+---------------+
| MaximumNumber |
+---------------+
|            78 |
+---------------+
1 row in set
mysql> CALL find_maximum(100, 50, 75);
+---------------+
| MaximumNumber |
+---------------+
|           100 |
+---------------+
1 row in set
mysql> CALL find_maximum(30, 30, 20);
+---------------+
| MaximumNumber |
+---------------+
|            30 |
+---------------+
1 row in set
```

Q6. Employee Salary Analysis

Create salary_analysis accepting employee name and salary. Allowance: salary >= 100,000 High/20%; salary >= 50,000 Medium/15%; otherwise Low/10%. Display employee name, salary, category, allowance percentage, allowance amount, and final salary.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE salary_analysis(IN ename VARCHAR(50), IN sal DECIMAL(12,2))
    -> BEGIN
    ->   DECLARE category_name VARCHAR(10);
    ->   DECLARE pct DECIMAL(5,2);
    ->   IF sal >= 100000 THEN SET category_name='High'; SET pct=20;
    ->   ELSEIF sal >= 50000 THEN SET category_name='Medium'; SET pct=15;
    ->   ELSE SET category_name='Low'; SET pct=10;
    ->   END IF;
    ->   SELECT ename AS EmployeeName, sal AS Salary, category_name AS Category,
    ->          pct AS AllowancePercent, sal*pct/100 AS Allowance,
    ->          sal+(sal*pct/100) AS FinalSalary;
    -> END //
Query OK, 0 rows affected
mysql> DELIMITER ;
mysql> CALL salary_analysis('Amit', 120000);
+--------------+-----------+----------+------------------+-----------+-------------+
| EmployeeName | Salary    | Category | AllowancePercent | Allowance | FinalSalary |
+--------------+-----------+----------+------------------+-----------+-------------+
| Amit         | 120000.00 | High     |            20.00 |  24000.00 |   144000.00 |
+--------------+-----------+----------+------------------+-----------+-------------+
1 row in set
mysql> CALL salary_analysis('Priya', 60000);
+--------------+----------+----------+------------------+-----------+-------------+
| EmployeeName | Salary   | Category | AllowancePercent | Allowance | FinalSalary |
+--------------+----------+----------+------------------+-----------+-------------+
| Priya        | 60000.00 | Medium   |            15.00 |   9000.00 |    69000.00 |
+--------------+----------+----------+------------------+-----------+-------------+
1 row in set
mysql> CALL salary_analysis('Rahul', 35000);
+--------------+----------+----------+------------------+-----------+-------------+
| EmployeeName | Salary   | Category | AllowancePercent | Allowance | FinalSalary |
+--------------+----------+----------+------------------+-----------+-------------+
| Rahul        | 35000.00 | Low      |            10.00 |   3500.00 |    38500.00 |
+--------------+----------+----------+------------------+-----------+-------------+
1 row in set
```

Q7. Electricity Bill Calculation

Create calculate_electricity_bill accepting consumer name and units consumed. Slab-wise rates: first 100 units Rs. 2/unit, next 100 Rs. 3/unit, next 200 Rs. 5/unit, above 400 Rs. 7/unit. Display consumer name, units consumed, and total bill.

```
mysql> DELIMITER //
mysql> CREATE PROCEDURE calculate_electricity_bill(IN cname VARCHAR(50), IN units INT)
    -> BEGIN
    ->   DECLARE bill DECIMAL(10,2) DEFAULT 0;
    ->   IF units <= 100 THEN
    ->     SET bill = units * 2;
    ->   ELSEIF units <= 200 THEN
    ->     SET bill = 100*2 + (units-100)*3;
    ->   ELSEIF units <= 400 THEN
    ->     SET bill = 100*2 + 100*3 + (units-200)*5;
    ->   ELSE
    ->     SET bill = 100*2 + 100*3 + 200*5 + (units-400)*7;
    ->   END IF;
    ->   SELECT cname AS ConsumerName, units AS UnitsConsumed, bill AS TotalBill;
    -> END //
Query OK, 0 rows affected
mysql> DELIMITER ;
mysql> CALL calculate_electricity_bill('Amit', 80);
+--------------+---------------+-----------+
| ConsumerName | UnitsConsumed | TotalBill |
+--------------+---------------+-----------+
| Amit         |            80 |    160.00 |
+--------------+---------------+-----------+
1 row in set
mysql> CALL calculate_electricity_bill('Priya', 150);
+--------------+---------------+-----------+
| ConsumerName | UnitsConsumed | TotalBill |
+--------------+---------------+-----------+
| Priya        |           150 |    350.00 |
+--------------+---------------+-----------+
1 row in set
mysql> CALL calculate_electricity_bill('Rahul', 350);
+--------------+---------------+-----------+
| ConsumerName | UnitsConsumed | TotalBill |
+--------------+---------------+-----------+
| Rahul        |           350 |   1250.00 |
+--------------+---------------+-----------+
1 row in set
mysql> CALL calculate_electricity_bill('Sneha', 500);
+--------------+---------------+-----------+
| ConsumerName | UnitsConsumed | TotalBill |
+--------------+---------------+-----------+
| Sneha        |           500 |   2200.00 |
+--------------+---------------+-----------+
1 row in set
```